

NVIDIA Corporation

Unique growth at very reasonable valuation, ignore quarterly noise

Vivek Arya

vivek.arya@bofa.com

Research Analyst

BofAS

Duksan Jang

duksan.jang@bofa.com

Research Analyst

BofAS

Michael Mani

michael.mani@bofa.com

Research Analyst

BofAS

28 August 2024 | Equity | United States | Semiconductors

NVDA US

Reiterate Rating	Price	Price Objective	Upside	Market Cap	Average Daily Value
BUY	125.61	165.00 ▲	31.4%	3,125,177	44047.93
	USD	from 150.00 USD		USD(mn)	USD(mn)
2025E EPS	2026E EPS	2027E EPS			
2.81 ▲	3.90 ▲	4.72 ▲			
from 2.58 USD	from 3.59 USD	from 4.44 USD			

all data as of 28 August 2024

Key takeaways

- Reit. Buy, top sector pick, raise ests/PO to \$165. N-t volatility could persist but I-t remains key genAI cycle beneficiary
- Likes: Hopper growth/strength to continue into Q3/Q4, sovereign AI demand, Ethernet products, strong purchase commitment
- Risks: Uncertainty around Blackwell ramp timing, GM headwinds from product startup costs, China data center contribution

See inside for Glossary

Q2 solid, Q3 good not great on Blackwell push-out

We reiterate our Buy rating on Nvidia (NVDA), a top sector pick, raising FY25/26E (roughly CY24/25E) pf-EPS by 9%/9% to \$2.81/\$3.90, and raising our PO to \$165 from \$150 on an unchanged 42x CY25E PE. It was a solid Q2 with sales of \$30bn vs. \$28.6bn consensus, however the stock is likely to be volatile as the Q3 sales outlook of \$32.5bn is only modestly ahead of \$31.9bn consensus, and below some more bullish \$33-\$35bn expectations, likely on Blackwell ramp pushed out by a quarter. We also flag rising Blackwell ramp costs n-t, with GM likely to be closer to 73% in Q4 vs 75% in Q3E. Despite the quarterly noise, we continue to believe in NVDA's unique growth opportunity, execution and dominant 80%+ share as generative AI deployments are still in their first 1-1.5yr of what is at least a 3-4 year upfront investment cycle.

Importantly, next-gen AI models will require 10x-20x more compute power to train (Blackwell only 3x-4x more compute over Hopper). AI deployment remains a mission-critical imperative for global cloud/enterprise customers, with NVDA providing the best turnkey model. Second, we highlight compelling valuation at 30-35x CY25E PE or <1x PE to growth (PEG) ratio relative to 40%+ EPS growth expectations, a standout in not just semis but also in large-cap tech/growth.

Likes: Hopper strength, sovereign, purchase commitments

1) Despite its 2+ year vintage, demand for Hopper remains solid, supporting the argument for both AI buildouts and for NVDA's product strength. Mgmt did not guide to Q4, but suggested demand for Hopper could be higher in 2H, with Blackwell contributing an incremental few \$bn. 2) NVDA flagged the rising demand from sovereign customers, raising the profile from high-single-digit \$bn to low-double-digit \$bn. Overall customer make-up in Q2 was a healthy mix of hyperscaler (45%) and internet service companies/ enterprise (50%). 3) Demand for AI Ethernet products remains strong with NVDA suggesting multi-billion \$ product line within a year. 4) Purchase commitments, while an imperfect sign, are still directionally positive, up 48% QoQ/149% YoY to \$27.8bn.

Risks: Blackwell ramp uncertainty, GM headwinds, China

1) Uncertainties around Blackwell ramp timing remains, related to supply-related headwinds. 2) Initial ramp of Blackwell could pressure GMs given its subscale volume and lower yield vs. more mature Hopper. 3) China revenue contribution grew to 12% in Q2, up 260bps QoQ, as H2O ramped in the quarter, though lower than prior 20%+ levels.

Estimates & Valuation

Estimates (Jan) (US\$)	2023A	2024A	2025E	2026E	2027E
EPS	0.33	1.30	2.81	3.90	4.72
GAAP EPS	0.17	1.19	2.67	3.66	4.46
EPS Change (YoY)	-25.0%	293.9%	116.2%	38.8%	21.0%
Consensus EPS (Bloomberg)			2.72	3.80	NA
DPS	0.02	0.02	0.02	0.02	0.02
Valuation (Jan)					
P/E	380.6x	96.6x	44.7x	32.2x	26.6x
GAAP P/E	738.9x	105.6x	47.0x	34.3x	28.2x
EV / EBITDA*	295.1x	80.8x	37.2x	27.0x	22.9x
Free Cash Flow Yield*	0.1%	0.9%	2.0%	2.7%	3.4%

Quarterly Estimates

Quarterly Earnings Estimates		
	2024	2025
Q1	0.11A	0.61A
Q2	0.27A	0.68A
Q3	0.40A	0.73E
Q4	0.52A	0.78E

Key Changes

(US\$)	Previous	Current
Price Obj.	150.00	165.00
2025E Rev (m)	114,044.5	123,895.8
2026E Rev (m)	156,425.4	171,344.3
2027E Rev (m)	185,832.7	201,064.9
2025E EPS	2.58	2.81
2026E EPS	3.59	3.90
2027E EPS	4.44	4.72

Stock Data

Price	125.61 USD
Price Objective	165.00 USD

Date Established	28-Aug-2024
Investment Opinion	C-1-7
52-Week Range	39.23 USD - 140.76 USD
Mrkt Val (mn) / Shares Out (mn)	3,125,177 USD / 24,880.0
Free Float	96.1%
Average Daily Value (mn)	44047.93 USD
BofA Ticker / Exchange	NVDA / NAS
Bloomberg / Reuters	NVDA US / NVDA.O
ROE (2025E)	112.0%
Net Dbt to Eqty (Jan-2024A)	5.7%
Average Daily Volume	350,672,128

Accelerator Market Outlook

Exhibit 1: Accelerator Market TAM Outlook

We see global accelerator TAM to reach ~\$218bn by CY27E and ~\$323bn by CY30E

Accelerator TAM Analysis	CY20	CY21	CY22	CY23	CY24E	CY25E	CY26E	CY27E	CY28E	CY29E	CY30E
NVDA Data Center (\$mn)	\$6,696	\$10,613	\$15,005	\$47,525	\$108,417	\$153,075	\$179,943	\$204,587	\$227,198	\$255,449	\$286,117
NVDA Networking (\$mn)	\$1,631	\$2,820	\$3,538	\$8,727	\$15,158	\$21,372	\$25,123	\$30,688	\$34,080	\$38,317	\$42,918
NVDA Compute (\$mn)	\$5,065	\$7,793	\$11,467	\$38,798	\$93,260	\$131,703	\$154,820	\$173,899	\$193,118	\$217,132	\$243,199
Assumed NVDA Revenue Share	85%	85%	82%	87%	84%	82%	80%	80%	78%	77%	75%
Global Accelerator TAM (\$mn)	\$5,959	\$9,168	\$13,985	\$44,854	\$111,288	\$161,007	\$194,010	\$217,918	\$246,639	\$282,724	\$322,974
YoY	70%	54%	53%	221%	148%	45%	20%	12%	13%	15%	14%

Accelerated Server Calculation

NVDA Data Center GPU Units (k, Mercury/BofA)

GPU ASP (\$)

GPU Per AI Server

GPU Content/AI Server (\$)

NVDA AI Servers (k)

Assumed NVDA Server Unit Share

Total Accelerated Servers (k)

Global Server Shipments (mn, IDC/BofA)

Accelerated Server (%)

	1,543	2,473	2,595	2,594	4,324	4,999	5,598	6,416	7,270	8,341	9,533
	\$3,283	\$3,152	\$4,419	\$14,957	\$21,568	\$26,349	\$27,659	\$27,106	\$26,564	\$26,032	\$25,512
	4.0	4.0	3.9	2.4	3.5	4.2	4.4	4.6	4.8	5.1	5.3
	\$13,131	\$12,607	\$17,357	\$36,344	\$75,404	\$110,424	\$122,835	\$124,444	\$128,596	\$132,532	\$136,260
	386	618	661	1,068	1,237	1,193	1,260	1,397	1,502	1,638	1,785
	70%	70%	68%	58%	53%	43%	40%	40%	40%	40%	40%
	551	883	966	1,841	2,334	2,806	3,151	3,494	3,754	4,096	4,462
	12.7	13.5	14.9	12.0	13.0	14.2	15.5	16.7	17.2	18.1	19.0
	4.4%	6.5%	6.5%	15.4%	17.9%	19.8%	20.3%	20.9%	21.8%	22.6%	23.5%

Cloud Servers as % of total servers

Total Cloud Servers (mn)

Accelerator Exposure to Cloud Customers

Implied Cloud Accelerated Servers (mn)

Implied Cloud Accelerator Attach-rate

	35%	35%	35%	35%	36%	37%	38%	38%	39%	39%	40%
	4.4	4.7	5.2	4.2	4.7	5.2	5.9	6.3	6.7	7.1	7.6
	65%	65%	68%	70%	71%	72%	73%	73%	74%	74%	75%
	0.4	0.6	0.7	1.3	1.7	2.0	2.3	2.6	2.8	3.0	3.3
	8.1%	12.1%	12.6%	30.7%	35.4%	38.5%	39.0%	40.2%	41.4%	43.0%	44.0%

Source: BofA Global Research estimates, Mercury Research, IDC

BofA GLOBAL RESEARCH

Exhibit 2: Accelerator Market Share Outlook

We expect Nvidia to maintain 75-80%+ share going forward

AI Accelerator Market Share	CY22	CY23	CY24E	CY25E	CY26E	CY27E
Nvidia (\$bn)	\$11.47	\$38.80	\$93.26	\$131.70	\$154.82	\$173.90
YoY		238%	140%	41%	18%	12%
Implied Share		82%	87%	84%	82%	80%
AMD (Consensus) (\$bn)	\$0.12	\$0.51	\$5.15	\$9.58	\$12.19	\$16.15
YoY		307%	918%	86%	27%	33%
Implied Share		1%	1%	5%	6%	7%
Broadcom (\$bn)	\$2.34	\$4.10	\$8.99	\$11.76	\$13.65	\$15.70
YoY		75%	119%	31%	16%	15%
Implied Share		17%	9%	8%	7%	7%
Marvell (\$bn)	\$0.00	\$0.04	\$0.60	\$1.11	\$1.28	\$1.38
YoY			1400%	85%	15%	8%
Implied Share		0%	0%	1%	1%	1%
Others (\$bn)	\$0.05	\$1.41	\$3.29	\$6.85	\$12.07	\$10.79
YoY		2571%	133%	108%	76%	-11%
Implied Share		0%	3%	3%	4%	5%
Total Market (\$bn)	\$13.98	\$44.85	\$111.29	\$161.01	\$194.01	\$217.92

YoY	221%	148%	45%	20%	12%
Source: BofA Global Research estimates, Mercury Research, IDC					
BofA GLOBAL RESEARCH					

Summary of model changes

Exhibit 3: Summary of estimate changes
We raise our FY25-27E sales by 8%-10%, EPS by 6%-9%

	Sales (\$mn)			EPS (Non GAAP, ex. options)		
	Old	NEW	delta	Old	NEW	delta
Apr-24	\$26,044	\$26,044	\$0	\$0.61	\$0.61	\$0.00
Jul-24	\$28,002	\$30,040	\$2,038	\$0.63	\$0.68	\$0.06
Oct-24E	\$30,089	\$32,516	\$2,427	\$0.67	\$0.73	\$0.06
Jan-25E	\$29,910	\$35,296	\$5,386	\$0.67	\$0.78	\$0.12
FY25E	\$114,044	\$123,896	\$9,851	\$2.58	\$2.81	\$0.23
YoY%	87.2%	103.4%	8.6%	98.8%	116.6%	9.0%
Apr-25E	\$33,837	\$37,634	\$3,797	\$0.77	\$0.84	\$0.07
Jul-25E	\$37,717	\$41,334	\$3,617	\$0.86	\$0.93	\$0.07
Oct-25E	\$40,177	\$43,883	\$3,706	\$0.92	\$1.00	\$0.08
Jan-26E	\$44,694	\$48,493	\$3,799	\$1.05	\$1.13	\$0.08
FY26E	\$156,425	\$171,344	\$14,919	\$3.59	\$3.90	\$0.31
YoY%	37.2%	38.3%	9.5%	39.4%	38.9%	8.6%
Apr-26E	\$45,642	\$49,288	\$3,645	\$1.07	\$1.15	\$0.07
Jul-26E	\$45,597	\$49,374	\$3,777	\$1.08	\$1.15	\$0.07
Oct-26E	\$45,400	\$49,323	\$3,923	\$1.08	\$1.16	\$0.07
Jan-27E	\$49,193	\$53,080	\$3,887	\$1.20	\$1.27	\$0.07
FY27E	\$185,833	\$201,065	\$15,232	\$4.44	\$4.72	\$0.28
YoY%	18.8%	17.3%	8.2%	23.6%	21.1%	6.4%
Source: BofA Global Research estimates, company reports						
BofA GLOBAL RESEARCH						

Q2 reported vs. expectations

Exhibit 4: NVDA Q2 results vs. Street expectations
Q2 sales came in 4% above initial expectations

	Jul-23	Apr-24	Jul-24						
			Original Guidance	Consensus	BofA	Reported	Delta vs. Street	Delta vs. Street (\$)	QoQ YoY
Sales (\$mn)	\$13,507	\$26,044	\$28,000 (+/- 2%)	\$28,857	\$28,002	\$30,040	4.1%	\$1,183	15.3% 122.4%
QoQ%	87.8%	17.8%		10.8%	7.5%	15.3%			
YoY%	101.5%	262.1%		113.6%	107.3%	122.4%			
Gaming	2,486	2,647		2,783	2,764	2,880	3%	\$97	9% 16%
Pro visualization (Quadro)	379	427		446	457	454	2%	\$8	6% 20%
Data center (Cloud, HPC, GRID)	10,323	22,563		25,165	24,373	26,272	4%	\$1,107	16% 154%
Automotive	253	329		350	326	346	-1%	(\$4)	5% 37%
OEM (incl. Crypto)	66	78		79	82	88	12%	\$10	13% 33%
Gross Margin % (GAAP)	70.1%	78.4%	74.8% +/- 50bp		74.8%	75.1%			
Gross Profit (Non GAAP, ex options)	9,614	20,560		\$21,776	21,145	22,729	4%	\$953	11% 136%
Gross Margin % (Non GAAP, ex options)	71.2%	78.9%	75.5% +/- 50bp	75.5%	75.5%	75.7%	20	-328	448
R&D (GAAP)	2,040	2,720			3,090	3,090			14% 51%
SG&A (GAAP)	622	777			932	842			8% 35%
Opex Total (GAAP)	2,662	3,497	\$4,000		4,022	3,932			12% 48%
Opex Total (Non GAAP, ex options)	1,838	2,501	\$2,800	\$2,922	2,802	2,792	-4%	(\$130)	12% 52%
Op. Income (Non GAAP, ex options)	7,776	18,059		\$18,854	18,343	19,937	6%	\$1,083	10% 156%
Op Margin % (Non GAAP, ex options)	57.6%	69.3%		65.3%	65.5%	66.4%	103	-297	880
Interest/other income (Non GAAP)	120	302	300.0		301	380			
Tax rate (Non GAAP)	15%	17%	17% +/- 1%		17%	17%			-4% 19%
Net Income (Non GAAP, ex options)	6,740	15,238			15,473	16,952			11% 152%
Diluted EPS (Non GAAP, ex options)	\$0.27	\$0.61		\$0.65	\$0.63	\$0.68	6%	\$0.04	11% 153%
Diluted Shares	24,990	24,890			24,735	24,848			0% -1%
Source: BofA Global Research estimates, company report, Bloomberg, Visible Alpha									
BofA GLOBAL RESEARCH									

Q3 guidance vs. expectations

Exhibit 5: NVDA Q3 guidance vs. our/Street expectations
NVDA guided Q3 sales \$650mn above Street expectations

Guidance v. Expectations	Jul-24 Reported	BofA	Consensus	Oct-24E Guidance	New BofA
Sales (\$mn)	\$30,040	\$30,089	\$31,854	\$32,500 (+/- 2%)	\$32,516
QoQ%	15.3%	7.5%	10.4%		8.2%
Gaming	2,880	3,016	3,042		3,142
Pro visualization (Quadro)	454	493	471		490
Data center (Cloud, HPC, GRID)	26,272	26,129	27,984		28,404
Automotive	346	365	379		388
OEM (incl. Crypto)	88	86	81		92
Gross Margin % (GAAP)	75.1%	73.9%		74.4% +/- 50bp	74.4%
Gross Profit (Non GAAP, ex options)	22,729	22,450	\$23,892		24,396
Gross Margin % (Non GAAP, ex options)	75.7%	74.6%	75.0%	75.0% +/- 50bp	75.0%
R&D (GAAP)	3,090	3,183			3,399
SG&A (GAAP)	842	937			901
Opex Total (GAAP)	3,932	4,120		\$4,300	4,300
Opex Total (Non GAAP, ex options)	2,792	2,870	\$2,874	\$3,000	3,000
Op. Income (Non GAAP, ex options)	19,937	19,581	\$21,018		21,396
Op Margin % (Non GAAP, ex options)	66.4%	65.1%	66.0%		65.8%
Interest/other income (Non GAAP)	380	301		\$350	351
Tax rate (Non GAAP)	17%	17%		17% +/- 1%	17%
Net Income (Non GAAP, ex options)	16,952	16,500			18,050
Diluted EPS (pf, excl. options)	\$0.68	\$0.67	\$0.71		\$0.73
Diluted Shares	24,848	24,580			24,705
Source: BofA Global Research estimates, company report, Bloomberg, Visible Alpha					
BofA GLOBAL RESEARCH					

Balance sheet, cash flow snapshot

Exhibit 6: Summary of Balance Sheet and Cash Flow trends

Net cash increased significantly QoQ

Balance Sheet							
	Jul-23	Oct-23	Jan-24	Apr-24	Jul-24	QoQ	YoY
Total Cash	\$16,023	\$18,281	\$25,984	\$31,438	\$34,800	10.7%	117.2%
ST Debt	\$1,249	\$1,249	\$1,250	\$1,250	\$0		
LT Debt	\$9,497	\$8,457	\$8,459	\$8,460	\$8,461		
Total Debt	\$10,746	\$9,706	\$9,709	\$9,710	\$8,461		
Net Cash	\$5,277	\$8,575	\$16,275	\$21,728	\$26,339	21.2%	399.1%
Net Cash/Share	\$0.21	\$0.34	\$0.65	\$0.87	\$1.06	21.4%	402.0%
DSO	47.6	41.7	41.2	43.2	42.8		
DPO	43.4	45.9	46.2	43.8	44.9		
Inventory \$	\$4,319	\$4,779	\$5,282	\$5,864	\$6,675	13.8%	54.5%
Inventory Turns	3.7	4.0	4.0	3.8	4.5		
Inventory Days	97	92	91	95	82	(13.3)	-15.8
Cash Conversion days	101.6	88.2	85.7	94.3	79.5		
Cash Flow from Ops	\$6,348	\$7,333	\$11,499	\$15,345	\$14,489		
Capex	(\$289)	(\$278)	(\$253)	(\$369)	(\$977)		
Free cash flow	\$6,059	\$7,055	\$11,246	\$14,976	\$13,512		

Source: BofA Global Research estimates, company report, Bloomberg

BoFA GLOBAL RESEARCH

Glossary

AI: Artificial Intelligence
AMD: Advanced Micro Devices
ASIC: Application Specific Integrated Circuit
DC: Data center
genAI: generative AI
GPU: graphics processor
GTC: GPU tech conference
NVDA: Nvidia
TAM: Total Addressable Market

Table 1: NVDA - Income Statement (\$mn)

NVDA reported \$30.0bn in sales and \$0.68 in EPS

FY (Jan)	1Q Apr-23	2Q Jul-23	3Q Oct-23	4Q Jan-24	1Q Apr-24	2Q Jul-24	3QE Oct-24E	4QE Jan-25E	1QE Apr-25E	2QE Jul-25E	3QE Oct-25E	4QE Jan-26E	FY24 Jan-24	FY25E Jan-25E
Revenues	\$7,192	\$13,507	\$18,120	\$22,103	\$26,044	\$30,040	\$32,516	\$35,296	\$37,634	\$41,334	\$43,883	\$48,493	\$60,922	\$123,896
QoQ	19%	88%	34%	22%	18%	15%	8%	9%	7%	10%	6%	11%		
YoY	-13%	101%	206%	265%	262%	122%	79%	60%	45%	38%	35%	37%	126%	103%
Cost of Sales (GAAP)	2,544	4,045	4,720	5,312	5,638	7,466	8,324	9,642	10,436	11,260	11,864	12,958	16,621	31,070
Gross profit (GAAP)	4,648	9,462	13,400	16,791	20,406	22,574	24,192	25,654	27,198	30,074	32,019	35,536	44,301	92,826
Gross profit (Non-GAAP, ex options)	4,802	9,614	13,583	16,959	20,560	22,729	24,396	25,901	27,462	30,364	32,326	35,875	44,958	93,586
R&D Expense (GAAP)	1,875	2,040	2,294	2,465	2,720	3,090	3,399	3,501	3,536	3,890	3,948	4,106	8,674	12,710
SG&A Expense (GAAP)	633	622	689	711	777	842	901	910	946	1,060	1,081	1,103	2,655	3,430
Opex (GAAP)	2,508	2,662	2,983	3,176	3,497	3,932	4,300	4,411	4,482	4,949	5,029	5,209	11,329	16,140
Opex (Non-GAAP, ex options)	1,750	1,838	2,026	2,210	2,501	2,792	3,000	3,075	3,117	3,541	3,592	3,722	7,824	11,368
Operating Income (Non-GAAP, ex options)	3,052	7,776	11,557	14,749	18,059	19,937	21,396	22,826	24,344	26,823	28,734	32,153	37,134	82,218
Interest/other Income (Non-GAAP)	84	120	175	232	302	380	351	351	401	401	401	401	611	1,384
Pretax Income (Non-GAAP, ex options)	3,136	7,896	11,732	14,981	18,361	20,317	21,747	23,177	24,745	27,224	29,135	32,554	37,745	83,602
Tax expense (Non-GAAP)	423	1,156	1,712	2,142	3,123	3,365	3,697	3,940	4,207	4,628	4,953	5,534	5,433	14,125
Net Income (GAAP)	2,043	6,188	9,243	12,285	14,881	16,599	16,801	17,922	19,186	21,186	22,734	25,504	29,759	66,203
Net Income (Non-GAAP, incl. options)	2,033	5,994	9,137	11,971	14,368	15,955	16,958	18,115	19,392	21,411	22,973	25,769	29,135	65,396
Net Income (Non-GAAP, ex options)	2,713	6,740	10,020	12,839	15,238	16,952	18,050	19,237	20,539	22,596	24,182	27,020	32,312	69,477
Diluted EPS (GAAP)	\$0.08	\$0.25	\$0.37	\$0.49	\$0.60	\$0.67	\$0.68	\$0.73	\$0.79	\$0.87	\$0.94	\$1.06	\$1.19	\$2.67
Diluted EPS (Non-GAAP, incl. options)	\$0.08	\$0.24	\$0.37	\$0.48	\$0.58	\$0.64	\$0.69	\$0.74	\$0.79	\$0.88	\$0.95	\$1.07	\$1.17	\$2.64
Diluted EPS (Non-GAAP, ex options)	\$0.11	\$0.27	\$0.40	\$0.52	\$0.61	\$0.68	\$0.73	\$0.78	\$0.84	\$0.93	\$1.00	\$1.13	\$1.30	\$2.81
Diluted Shares	24,900	24,990	24,940	24,900	24,890	24,848	24,705	24,562	24,419	24,275	24,132	23,989	24,933	24,751
Dividend	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.02	\$0.02
Margins (Non-GAAP, ex options)														
Gross Margin	66.8%	71.2%	75.0%	76.7%	78.9%	75.7%	75.0%	73.4%	73.0%	73.5%	73.7%	74.0%	73.8%	75.5%
Opex	24.3%	13.6%	11.2%	10.0%	9.6%	9.3%	9.2%	8.7%	8.3%	8.6%	8.2%	7.7%	12.8%	9.2%
Operating Income	42.4%	57.6%	63.8%	66.7%	69.3%	66.4%	65.8%	64.7%	64.7%	64.9%	65.5%	66.3%	61.0%	66.4%
Tax rate	13.5%	14.6%	14.6%	14.3%	17.0%	16.6%	17.0%	17.0%	17.0%	17.0%	17.0%	17.0%	14.4%	16.9%
Net Income	37.7%	49.9%	55.3%	58.1%	58.5%	56.4%	55.5%	54.5%	54.6%	54.7%	55.1%	55.7%	53.0%	56.1%

Source: BofA Global Research estimates, company report

BoFA GLOBAL RESEARCH

Table 2: NVDA - Segment Model (\$mn)

NVDA FY23 sales on track to reach >\$100bn

FY (Jan)	1Q Apr-23	2Q Jul-23	3Q Oct-23	4Q Jan-24	1Q Apr-24	2Q Jul-24	3QE Oct-24E	4QE Jan-25E	1QE Apr-25E	2QE Jul-25E	3QE Oct-25E	4QE Jan-26E	FY24 Jan-24	FY25E Jan-25E	FY26E
Gaming	2,240	2,486	2,856	2,865	2,647	2,880	3,142	3,101	3,018	3,309	3,642	3,673	10,447	11,770	13,642
Pro visualization (Quadro)	295	379	416	463	427	454	490	485	476	499	504	515	1,553	1,857	1,994
Data center															
(Compute/Networking)	4,284	10,323	14,514	18,404	22,563	26,272	28,404	31,178	33,542	36,878	39,059	43,597	47,525	108,417	153,075
Compute	3,357	8,612	11,756	15,073	19,392	22,604	24,438	26,825	28,859	31,729	33,605	37,510	38,798	93,260	131,703
Networking	927	1,711	2,758	3,331	3,171	3,668	3,966	4,353	4,683	5,149	5,453	6,087	8,727	15,158	21,372
Automotive	296	253	261	281	329	346	388	446	499	549	576	605	1,091	1,508	2,230
OTM & Other	77	66	73	90	78	88	92	85	99	99	101	103	306	343	403
Total	7,192	13,507	18,120	22,103	26,044	30,040	32,516	35,296	37,634	41,334	43,883	48,493	60,922	123,896	171,344

QoQ%													
Gaming	22%	11%	15%	0%	-8%	9%	9%	-1%	-3%	10%	10%	1%	
Pro visualization (Quadro)	31%	28%	10%	11%	-8%	6%	8%	-1%	-2%	5%	1%	2%	
Data center													
(Compute/Networking)	18%	141%	41%	27%	23%	16%	8%	10%	8%	10%	6%	12%	
Compute	25%	157%	37%	28%	29%	17%	8%	10%	8%	10%	6%	12%	
Networking	-1%	85%	61%	21%	-5%	16%	8%	10%	8%	10%	6%	12%	
Automotive	1%	-15%	3%	8%	17%	5%	12%	15%	12%	10%	5%	5%	
QEM & Other	-8%	-14%	11%	23%	-13%	13%	5%	-8%	17%	0%	2%	2%	
Total	19%	88%	34%	22%	18%	15%	8%	9%	7%	10%	6%	11%	
YoY%													
Gaming	-38%	22%	81%	56%	18%	16%	10%	8%	14%	15%	16%	18%	15%
Pro visualization (Quadro)	-53%	-24%	108%	105%	45%	20%	18%	5%	11%	10%	3%	6%	1%
Data center													
(Compute/Networking)	14%	171%	279%	409%	427%	154%	96%	69%	49%	40%	38%	40%	217%
Compute	13%	193%	308%	462%	478%	162%	108%	78%	49%	40%	38%	40%	238%
Networking	17%	98%	191%	256%	242%	114%	44%	31%	48%	40%	38%	40%	147%
Automotive	114%	15%	4%	-4%	11%	37%	48%	59%	52%	59%	49%	36%	21%
QEM & Other	-51%	-53%	7%	2%	1%	33%	26%	-5%	27%	13%	10%	21%	-33%
Total	-13%	101%	206%	265%	262%	122%	79%	60%	45%	38%	35%	37%	126%
% of Sales													
Gaming	31%	18%	16%	13%	10%	10%	10%	9%	8%	8%	8%	8%	17%
Pro visualization (Quadro)	4%	3%	2%	2%	2%	2%	2%	1%	1%	1%	1%	1%	3%
Data center													
(Compute/Networking)	60%	76%	80%	83%	87%	87%	87%	88%	89%	89%	89%	90%	78%
Compute	47%	64%	65%	68%	74%	75%	75%	76%	77%	77%	77%	77%	64%
Networking	13%	13%	15%	15%	12%	12%	12%	12%	12%	12%	12%	13%	14%
Automotive	4%	2%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	2%
QEM & Other	1%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	1%
Total	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

Source: BofA Global Research estimates, company report

BofA GLOBAL RESEARCH

Exhibit 7: NVDA - Balance Sheet (\$mn)
We model net cash per share strongly expanding over the next several quarters

	1Q	2Q	3Q	4Q	1Q	2Q	3QE	4QE	1QE	2QE	3QE	4QE	FY24	FY25E	FY26E
FY (Jan)	Apr-23	Jul-23	Oct-23	Jan-24	Apr-24	Jul-24	Oct-24E	Jan-25E	Apr-25E	Jul-25E	Oct-25E	Jan-26E	Jan-24	Jan-25E	Jan-26E
Assets															
Cash & Equivalents	5,079	5,882	5,519	7,280	7,587	8,571	11,587	15,385	19,255	26,426	35,835	46,779	7,280	15,385	46,779
ST Investments	10,241	10,141	12,762	18,704	23,851	26,229	31,973	37,717	43,461	49,205	54,949	60,693	18,704	37,717	60,693
Accounts receivable	4,080	7,066	8,309	9,999	12,365	14,132	15,297	16,605	17,705	19,445	20,644	22,813	9,999	16,605	22,813
Inventory	4,611	4,319	4,779	5,282	5,864	6,675	7,442	8,621	10,865	11,722	12,351	13,490	5,282	8,621	13,490
Total current assets	24,883	28,797	32,658	44,345	53,729	59,633	70,325	82,353	95,312	110,824	127,806	147,801	44,345	82,353	147,801
PP&E	3,740	3,799	3,844	3,914	4,006	4,885	5,474	6,113	6,301	6,508	6,727	6,970	3,914	6,113	6,970
Goodwill	4,430	4,430	4,430	4,430	4,453	4,622	4,622	4,622	4,622	4,622	4,622	4,622	4,430	4,622	4,622
Intangible Assets	1,541	1,395	1,251	1,112	986	952	952	952	952	952	952	952	1,112	952	952
Other Assets	15,837	16,959	17,646	17,469	19,337	20,709	20,709	20,709	20,709	20,709	20,709	20,709	17,469	20,709	20,709
Total Assets	44,460	49,555	54,148	65,728	77,072	85,227	96,508	109,175	122,322	138,041	155,242	175,480	65,728	109,175	175,480
Liabilities & Shareholders Equity															
Accounts payable	1,141	1,929	2,380	2,699	2,715	3,680	4,103	4,753	4,587	4,949	5,215	5,696	2,699	4,753	5,696
Accrued liabilities	4,869	7,156	5,472	6,682	11,258	10,289	10,289	10,289	10,289	10,289	10,289	10,289	6,682	10,289	10,289
Total current liabilities	7,260	10,334	9,101	10,631	15,223	13,969	14,392	15,042	14,876	15,238	15,504	15,985	10,631	15,042	15,985
Total non-current liabilities	12,680	11,720	11,782	12,119	12,707	13,101	13,101	13,101	13,101	13,101	13,101	13,101	12,119	13,101	13,101
Total shareholders' equity	24,520	27,501	33,265	42,978	49,142	58,157	69,015	81,033	94,345	109,702	126,637	146,394	42,978	81,033	146,394
Total Liab. & Stock Equity	44,460	49,555	54,148	65,728	77,072	85,227	96,508	109,175	122,322	138,041	155,242	175,480	65,728	109,175	175,480
Cash and Liquidity Ratios															
Total Cash	15,320	16,023	18,281	25,984	31,438	34,800	43,560	53,102	62,716	75,631	90,784	107,472	25,984	53,102	107,472
Short-term debt	1,250	1,249	1,249	1,250	1,250	0	0	0	0	0	0	0	1,250	0	0
Long-term debt	9,704	9,497	8,457	8,459	8,460	8,461	8,461	8,461	8,461	8,461	8,461	8,461	8,459	8,461	8,461
Total debt	10,954	10,746	9,706	9,709	9,710	8,461	8,461	8,461	8,461	8,461	8,461	8,461	9,709	8,461	8,461
Net Cash	4,366	5,277	8,575	16,275	21,728	26,339	35,099	44,641	54,255	67,170	82,323	99,011	16,275	44,641	99,011
Net cash / share	\$0.18	\$0.21	\$0.34	\$0.65	\$0.87	\$1.06	\$1.42	\$1.82	\$2.22	\$2.77	\$3.41	\$4.13	\$0.65	\$1.80	\$4.09
Cash conversion ratios															
Days Sales Outstanding (DSO)	51.6	47.6	41.7	41.2	43.2	42.8	42.8	42.8	42.8	42.8	42.8	42.8	42.8	42.8	42.8
Payable Days (DPO)	40.8	43.4	45.9	46.2	43.8	44.9	44.9	44.9	40.0	40.0	40.0	40.0	40.0	40.0	40.0
Inventory Turns	2.2	3.7	4.0	4.0	3.8	4.5	4.5	4.5	3.8	3.8	3.8	3.8	3.8	3.8	3.8
Inventory Days (DIO)	165.4	97.4	92.4	90.7	94.9	81.6	81.6	81.6	95.0	95.0	95.0	95.0	95.0	95.0	95.0
Cash Conversion Days	176.2	101.6	88.2	85.7	94.3	79.5	79.5	79.5	97.8	97.8	97.8	97.8			

Source: BofA Global Research, company reports

BofA GLOBAL RESEARCH

Table 3: NVDA - Cash flow Statement (\$mn)
We see path towards 50%+ FCF margin long-term

	1Q	2Q	3Q	4Q	1Q	2Q	3QE	4QE	1QE	2QE	3QE	4QE	FY24	FY25E	FY26E
FY (Jan)	Apr-23	Jul-23	Oct-23	Jan-24	Apr-24	Jul-24	Oct-24E	Jan-25E	Apr-25E	Jul-25E	Oct-25E	Jan-26E	Jan-24	Jan-25E	Jan-26E
Operating Activities - Cash Flow															
Net income	2,043	6,188	9,243	12,285	14,881	16,599	16,801	17,922	19,186	21,186	22,734	25,504	29,759	66,203	88,609
Depreciation	384	365	372	387	410	433	469	509	753	827	878	970	1,508	1,820	3,427
Amortization of Intangibles	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Stock Compensation Expenses	735	842	979	993	1,011	1,154	1,315	1,352	1,382	1,427	1,457	1,508	3,549	4,832	5,774
Deferred income taxes	(1,135)	(746)	(529)	(78)	(1,577)	(1,699)	0	0	0	0	0	0	(2,488)	(3,276)	0
Patents License Payments	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Other non-cash activity	(20)	(129)	3	(369)	(214)	(337)	0	0	0	0	0	0	(515)	(551)	0
Change In:															
Accounts Receivables	(252)	(2,986)	(1,243)	(1,690)	(2,366)	(1,767)	(1,165)	(1,308)	(1,100)	(1,740)	(1,199)	(2,169)	(6,171)	(6,606)	(6,209)
Inventories	566	296	(457)	(503)	(577)	(803)	(767)	(1,178)	(2,244)	(858)	(629)	(1,139)	(98)	(3,326)	(4,870)
Prepaid Expenses/Other	(215)	(376)	254	(1,184)	(726)	714	0	0	0	0	0	0	(1,521)	(12)	0
Accounts Payable	11	777	461	281	(22)	823	423	650	(165)	362	266	481	1,530	1,874	943
Accrued Liabilities	689	1,986	(1,722)	1,072	4,202	(888)	0	0	0	0	0	0	2,025	3,314	0
Cash From Operations	2,911	6,348	7,333	11,499	15,345	14,489	17,075	17,946	17,812	21,204	23,505	25,154	28,091	64,856	87,675
Investing Activities - Cash Flow															
Capex	(248)	(289)	(278)	(253)	(369)	(977)	(1,058)	(1,148)	(941)	(1,033)	(1,097)	(1,212)	(1,068)	(3,551)	(4,284)
ST Investment Purchases	(2,801)	(2,542)	(5,345)	(7,524)	(9,303)	(5,744)	(5,744)	(5,744)	(5,744)	(5,744)	(5,744)	(5,744)	(18,212)	(26,535)	(22,976)

ST Investment Maturities	2,512	2,598	2,891	1,781	4,153	4,109	0	0	0	0	0	0	0	9,782	8,262	0
Cash From Investing	(841)	(447)	(3,169)	(6,109)	(5,693)	(3,185)	(6,802)	(6,892)	(6,685)	(6,777)	(6,841)	(6,956)		(10,566)	(22,571)	(27,260)
Financing Activities - Cash Flow																
Stock Repurchases	0	(3,067)	(3,806)	(2,660)	(7,740)	(7,158)	(7,158)	(7,158)	(7,158)	(7,158)	(7,158)	(7,158)		(9,533)	(29,214)	(28,632)
Common Stock Issuance	0	0	0	(29)	0	0	0	0	0	0	0	0		0	0	0
Proceeds from Emp. Stock Plans	246	1	156	0	285	0	0	0	0	0	0	0		403	285	0
Capital Obligations Payments	0	0	0	(29)	0	0	0	0	0	0	0	0		(29)	0	0
Cash From Financing	(380)	(5,098)	(4,527)	(3,629)	(9,345)	(10,320)	(7,257)	(7,257)	(7,256)	(7,256)	(7,255)	(7,255)		(13,634)	(34,179)	(29,022)
Net Change in Cash/Equ.	1,690	803	(363)	1,761	307	984	3,016	3,798	3,871	7,171	9,409	10,943		3,891	8,105	31,394
Cash/Equ. at beginning of period	3,389	5,079	5,882	5,519	7,280	7,587	8,571	11,587	15,385	19,255	26,426	35,835		3,389	7,280	15,385
Cash/Equ. at end of period	5,079	5,882	5,519	7,280	7,587	8,571	11,587	15,385	19,255	26,426	35,835	46,779		7,280	15,385	46,779
Free Cash Flow	2,663	6,059	7,055	11,246	14,976	13,512	16,018	16,798	16,871	20,170	22,408	23,942		27,023	61,304	83,391
Trailing 4Q FCF	5,102	10,324	17,517	27,023	39,336	46,789	55,752	61,304	63,199	69,857	76,248	83,391				
4Q FCF/Dil Share	\$0.20	\$0.41	\$0.70	\$1.09	\$1.58	\$1.88	\$2.26	\$2.50	\$2.59	\$2.88	\$3.16	\$3.48		\$1.08	\$2.48	\$3.45
Source: BofA Global Research estimates, company report																
BofA GLOBAL RESEARCH																

BofA Securities does and seeks to do business with issuers covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.

Click for [important disclosures](#), [Analyst Certification](#), [Price Objective Basis & Risk](#).

iQprofileSM NVIDIA Corporation

Company Description

Nvidia designs and sell graphics and video processing chips for desktop and notebook gaming PCs, workstations, game consoles, and accelerated computing servers and supercomputers.

Investment Rationale

Our positive view on Nvidia is based on its underappreciated transformation from a traditional PC graphics chip vendor, into a supplier into high-end gaming, enterprise graphics, cloud, accelerated computing and automotive markets. The company has executed consistently and has a solid balance sheet with demonstrated commitment to capital returns.

iQmethod SM - Bus Performance*

(US\$ Millions)	2023A	2024A	2025E	2026E	2027E
Return on Capital Employed	21.5%	70.9%	92.4%	73.3%	54.8%
Return on Equity	34.3%	99.3%	112.0%	83.0%	59.5%
Operating Margin	33.5%	61.0%	66.4%	65.4%	66.0%
Free Cash Flow	3,809	27,023	61,304	83,391	107,114

iQmethod SM - Quality of Earnings*

(US\$ Millions)	2023A	2024A	2025E	2026E	2027E
Cash Realization Ratio	0.7x	0.9x	0.9x	0.9x	1.0x
Asset Replacement Ratio	1.2x	0.7x	2.0x	1.3x	1.2x
Tax Rate	NM	12.0%	15.5%	17.0%	17.0%
Net Debt-to-Equity Ratio	34.2%	5.7%	-8.5%	-26.2%	-40.8%
Interest Cover	NA	NA	NA	NA	NA

Income Statement Data (Jan)

(US\$ Millions)	2023A	2024A	2025E	2026E	2027E
Sales	26,974	60,922	123,896	171,344	201,065
% Change	0.2%	125.9%	103.4%	38.3%	17.3%
Gross Profit	15,965	44,958	93,586	126,027	149,691
% Change	-11.1%	181.6%	108.2%	34.7%	18.8%
EBITDA	10,584	38,642	84,038	115,482	136,650
% Change	-23.7%	265.1%	117.5%	37.4%	18.3%
Net Interest & Other Income	6	611	1,384	1,604	1,804
Net Income (Adjusted)	8,366	32,312	69,477	94,337	111,579
% Change	-25.7%	286.2%	115.0%	35.8%	18.3%

Free Cash Flow Data (Jan)

(US\$ Millions)	2023A	2024A	2025E	2026E	2027E
Net Income from Cont Operations (GAAP)	4,368	29,759	66,203	88,609	105,413
Depreciation & Amortization	1,544	1,508	1,820	3,427	4,021
Change in Working Capital	(2,206)	(3,722)	(4,173)	(10,135)	(3,398)
Deferred Taxation Change	(2,164)	(2,488)	(3,276)	0	0
Other Adjustments, Net	4,100	3,034	4,281	5,774	6,104
Capital Expenditure	(1,833)	(1,068)	(3,551)	(4,284)	(5,027)
Free Cash Flow	3,809	27,023	61,304	83,391	107,114
% Change	-53.2%	609.5%	126.9%	36.0%	28.4%

Share / Issue Repurchase	(9,685)	(9,130)	(28,929)	(28,632)	(28,632)
Cost of Dividends Paid	(398)	(396)	(542)	(390)	(380)
Change in Debt	(22)	(1,279)	(1,250)	0	0

Balance Sheet Data (Jan)
(US\$ Millions)

	2023A	2024A	2025E	2026E	2027E
Cash & Equivalents	3,389	7,280	15,385	46,779	101,904
Trade Receivables	3,827	9,999	16,605	22,813	24,971
Other Current Assets	15,857	27,066	50,364	78,209	102,781
Property, Plant & Equipment	3,807	3,914	6,113	6,970	7,975
Other Non-Current Assets	14,302	17,469	20,709	20,709	20,709
Total Assets	41,182	65,728	109,175	175,480	258,339
Short-Term Debt	1,250	1,250	0	0	0
Other Current Liabilities	5,313	9,381	15,042	15,985	16,340
Long-Term Debt	9,703	8,459	8,461	8,461	8,461
Other Non-Current Liabilities	2,815	3,660	4,640	4,640	4,640
Total Liabilities	19,081	22,750	28,143	29,086	29,441
Total Equity	22,101	42,978	81,033	146,394	228,899
Total Equity & Liabilities	41,182	65,728	109,175	175,480	258,339

* Click for full definitions of iQmethod™ measures.

Price Objective Basis & Risk

NVIDIA Corporation (NVDA)

Our \$165 PO is based on 42x CY25E PE ex cash, within NVDA's historical 26x-69x forward year PE range, which we believe is justified given stronger growth opportunities ahead as gaming cycle troughs and data center demand potentially faces strong, long-term demand dynamics.

Downside risks are: 1) weakness in consumer driven gaming market, 2) Competition with major public firms, internal cloud projects and other private companies in accelerated computing markets, 3) Larger than expected impact from restrictions on compute shipments to China, or additional restrictions placed on activity in the region, 4) Lumpy and unpredictable sales in new enterprise, data center, and autos markets, 5) Potential for decelerating capital returns.

Coverage Cluster

US - Semiconductors and Semiconductor Capital Equipment Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY	Advanced Micro Devices, Inc.	AMD	AMD US	Vivek Arya
	Analog Devices Inc.	ADI	ADI US	Vivek Arya
	Applied Materials, Inc.	AMAT	AMAT US	Vivek Arya
	Arm Holdings	ARM	ARM US	Vivek Arya
	Broadcom Inc.	AVGO	AVGO US	Vivek Arya
	Cadence	CDNS	CDNS US	Vivek Arya
	Camtek	CAMT	CAMT US	Vivek Arya
	Coherent Corp.	COHR	COHR US	Vivek Arya
	KLA Corporation	KLAC	KLAC US	Vivek Arya
	Lam Research Corp.	LRCX	LRCX US	Vivek Arya
	M/A-Com	MTSI	MTSI US	Vivek Arya
	Marvell Technology, Inc.	MIRVL	MIRVL US	Vivek Arya
	Micron Technology, Inc.	MU	MU US	Vivek Arya
	MKS Instruments	MKSI	MKSI US	Vivek Arya
	Nova	NVMI	NVMI US	Vivek Arya
	NVIDIA Corporation	NVDA	NVDA US	Vivek Arya
	NXP Semiconductors NV	NXPI	NXPI US	Vivek Arya
	onsemi	ON	ON US	Vivek Arya
	Synopsys	SNPS	SNPS US	Vivek Arya
NEUTRAL	Advanced Energy Industries	AEIS	AEIS US	Duksan Jang
	Axcelis Technologies	ACLS	ACLS US	Duksan Jang
	GlobalFoundries	GFS	GFS US	Vivek Arya
	Microchip	MCHP	MCHP US	Vivek Arya
	Texas Instruments Inc.	TXN	TXN US	Vivek Arya
UNDERPERFORM	Ambarella	AMBA	AMBA US	Vivek Arya
	Credo Technology	CRDO	CRDO US	Vivek Arya
	Intel	INTC	INTC US	Vivek Arya
	Lattice Semiconductor	LSCC	LSCC US	Duksan Jang
	Lumentum Holdings	LITE	LITE US	Vivek Arya
	Qorvo Inc.	QRVO	QRVO US	Vivek Arya
	Skyworks Solutions, Inc.	SWKS	SWKS US	Vivek Arya
	Teradyne	TER	TER US	Vivek Arya
	Wolfspeed Inc.	WOLF	WOLF US	Vivek Arya

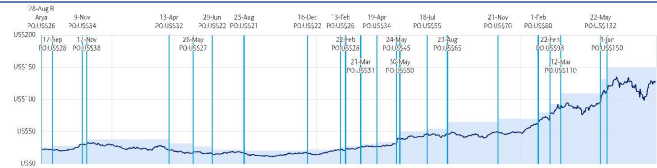
Analyst Certification

I, Vivek Arya, hereby certify that the views expressed in this research report accurately reflect my personal views about the subject securities and issuers. I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or view expressed in this research report.

Disclosures

Important Disclosures

NVIDIA (NVDA) Price Chart



The Investment Opinion System is contained at the end of the report, under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

Equity Investment Rating Distribution: Technology Group (as of 30 Jun 2024)

Coverage Universe	Count	Percent	Inv. Banking Relationships	Count	Percent
Buy	221	55.53%	Buy	114	51.58%
Hold	89	22.36%	Hold	45	50.56%
Sell	88	22.11%	Sell	23	26.14%

Equity Investment Rating Distribution: Global Group (as of 30 Jun 2024)

Coverage Universe	Count	Percent	Inv. Banking Relationships	Count	Percent
Buy	1857	54.79%	Buy	1084	58.37%
Hold	758	22.37%	Hold	446	58.84%
Sell	774	22.84%	Sell	350	45.22%

Issues that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.

FUNDAMENTAL EQUITY OPINION KEY: Opinions include a Volatility Risk Rating, an Investment Rating and an Income Rating. VOLATILITY RISK RATINGS, indicators of potential price fluctuation, are: A - Low, B - Medium and C - High. INVESTMENT RATINGS reflect the analyst's assessment of both a stock's absolute total return potential as well as its attractiveness for investment relative to other stocks within its Coverage Cluster (defined below). Our investment ratings are: 1 - Buy stocks are expected to have a total return of at least 10% and are the most attractive stocks in the coverage cluster; 2 - Neutral stocks are expected to remain flat or increase in value and are less attractive than Buy rated stocks and 3 - Underperform stocks are the least attractive stocks in a coverage cluster. An investment rating of 6 (No Rating) indicates that a stock is no longer trading on the basis of fundamentals. Analysts assign investment ratings considering, among other things, the 0-12 month total return expectation for a stock and the firm's guidelines for ratings dispersions (shown in the table below). The current price objective for a stock should be referenced to better understand the total return expectation at any given time. The price objective reflects the analyst's view of the potential price appreciation (depreciation).

Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster
Buy	≥ 10%	± 70%
Neutral	≥ 0%	± 30%
Underperform	N/A	± 20%

Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

INCOME RATINGS, indicators of potential cash dividends, are: 7 - same/higher (dividend considered to be secure), 8 - same/lower (dividend not considered to be secure) and 9 - pays no cash dividend. Coverage Cluster is comprised of stocks covered by a single analyst or two or more analysts sharing a common industry, sector, region or other classification(s). A stock's coverage cluster is included in the most recent BofA Global Research report referencing the stock.

Price Charts for the securities referenced in this research report are available on the Price Charts website (https://pricecharts.baml.com/), or call 1-800-MERRILL to have them mailed.

BofAS or one of its affiliates acts as a market maker for the equity securities recommended in the report: NVIDIA.

The issuer is or was, within the last 12 months, an investment banking client of BofAS and/or one or more of its affiliates: NVIDIA.

BofAS or an affiliate has received compensation from the issuer for non-investment banking services or products within the past 12 months: NVIDIA.

The issuer is or was, within the last 12 months, a non-securities business client of BofAS and/or one or more of its affiliates: NVIDIA.

BofAS or an affiliate has received compensation for investment banking services from this issuer within the past 12 months: NVIDIA.

BofAS together with its affiliates beneficially owns one percent or more of the common stock of this issuer. If this report was issued on or after the 9th day of the month, it reflects the ownership position on the last day of the previous month. Reports issued before the 9th day of a month reflect the ownership position at the end of the second month preceding the date of the report: NVIDIA.

BofAS or one of its affiliates is willing to sell to, or buy from, clients the common equity of the issuer on a principal basis: NVIDIA.

The issuer is or was, within the last 12 months, a securities business client (non-investment banking) of BofAS and/or one or more of its affiliates: NVIDIA.

BofA Global Research personnel (including the analyst(s) responsible for this report) receive compensation based upon, among other factors, the overall profitability of Bank of America Corporation, including profits derived from investment banking. The analyst(s) responsible for this report may also receive compensation based upon, among other factors, the overall profitability of the Bank's sales and trading businesses relating to the class of securities or financial instruments for which such analyst is responsible.

Other Important Disclosures

From time to time research analysts conduct site visits of covered issuers. BofA Global Research policies prohibit research analysts from accepting payment or reimbursement for travel expenses from the issuer for such visits.

Prices are indicative and for information purposes only. Except as otherwise stated in the report, for any recommendation in relation to an equity security, the price referenced is the publicly traded price of the security as of close of business on the day prior to the date of the report or, if the report is published during intraday trading, the price referenced is indicative of the traded price as of the date and time of the report and in relation to a debt security (including equity preferred and CDS), prices are indicative as of the date and time of the report and are from various sources including BofA Securities trading desks.

The date and time of completion of the production of any recommendation in this report shall be the date and time of dissemination of this report as recorded in the report timestamp.

Recipients who are not institutional investors or market professionals should seek the advice of their independent financial advisor before considering information in this report in connection with any investment decision, or for a necessary explanation of its contents.

Officers of BoFAS or one or more of its affiliates (other than research analysts) may have a financial interest in securities of the issuer(s) or in related investments.

Refer to [BoFA Global Research policies relating to conflicts of interest \(https://rsch.baml.com/col\)](https://rsch.baml.com/col).

"BoFA Securities" includes BoFA Securities, Inc. ("BoFAS") and its affiliates. Investors should contact their BoFA Securities representative or Merrill Global Wealth Management financial advisor if they have questions concerning this report or concerning the appropriateness of any investment idea described herein for such investor. "BoFA Securities" is a global brand for BoFA Global Research.

Information relating to Non-US affiliates of BoFA Securities and Distribution of Affiliate Research Reports:

BoFAS and/or Merrill Lynch, Pierce, Fenner & Smith Incorporated ("MLPF&S") may in the future distribute, information of the following non-US affiliates in the US (short name: legal name, regulator): Merrill Lynch (South Africa): Merrill Lynch South Africa (Pty) Ltd., regulated by The Financial Service Board; MLJ (UK): Merrill Lynch International, regulated by the Financial Conduct Authority (FCA) and the Prudential Regulation Authority (PRA); BoFASE (France): BoFA Securities Europe SA is authorized by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers (AMF). BoFA Securities Europe SA ("BoFASE") with registered address at 51, rue La Boétie, 75008 Paris is registered under no 842 602 690 RCS Paris. In accordance with the provisions of French Code Monétaire et Financier (Monetary and Financial Code), BoFASE is an établissement de crédit et d'investissement (credit and investment institution) that is authorised and supervised by the European Central Bank and the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers. BoFASE's share capital can be found at www.bofam.com/BoFASEdisclaimer (<http://www.bofam.com/BoFASEdisclaimer>); BoFA Europe (Milan): Bank of America Europe Designated Activity Company, Milan Branch, regulated by the Bank of Italy, the European Central Bank (ECB) and the Central Bank of Ireland (CBI); BoFA Europe (Frankfurt): Bank of America Europe Designated Activity Company, Frankfurt Branch regulated by BaFin, the ECB and the CBI; BoFA Europe (Madrid): Bank of America Europe Designated Activity Company, Sucursal en España, regulated by the Bank of Spain, the ECB and the CBI; Merrill Lynch (Australia): Merrill Lynch Equities (Australia) Limited, regulated by the Australian Securities and Investments Commission; Merrill Lynch (Hong Kong): Merrill Lynch (Asia Pacific) Limited, regulated by the Hong Kong Securities and Futures Commission (HKSF); Merrill Lynch (Singapore): Merrill Lynch (Singapore) Pte Ltd, regulated by the Monetary Authority of Singapore (MAS); Merrill Lynch (Canada): Merrill Lynch Canada Inc, regulated by the Canadian Investment Regulatory Organization; Merrill Lynch (Mexico): Merrill Lynch Mexico, SA de CV, Casa de Bolsa, regulated by the Comisión Nacional Bancaria y de Valores; BoFAS Japan: BoFA Securities Japan Co., Ltd., regulated by the Financial Services Agency; Merrill Lynch (Seoul): Merrill Lynch International, LLC Seoul Branch, regulated by the Financial Supervisory Service; Merrill Lynch (Taiwan): Merrill Lynch Securities (Taiwan) Ltd., regulated by the Securities and Futures Bureau; BoFAS India: BoFA Securities India Limited, regulated by the Securities and Exchange Board of India (SEBI); Merrill Lynch (Israel): Merrill Lynch Israel Limited, regulated by Israel Securities Authority; Merrill Lynch (DIFC): Merrill Lynch International (DIFC Branch), regulated by the Dubai Financial Services Authority (DFSA); Merrill Lynch (Brazil): Merrill Lynch S.A. Corretora de Títulos e Valores Mobiliários, regulated by Comissão de Valores Mobiliários; Merrill Lynch KSA Company: Merrill Lynch Kingdom of Saudi Arabia Company, regulated by the Capital Market Authority.

This information has been approved for publication and is distributed in the United Kingdom (UK) to professional clients and eligible counterparties (as each is defined in the rules of the FCA and the PRA) by MLJ (UK), which is authorized by the PRA and regulated by the FCA and the PRA - details about the extent of our regulation by the FCA and PRA are available from us on request; has been approved for publication and is distributed in the European Economic Area (EEA) by BoFASE (France), which is authorized by the ACPR and regulated by the ACPR and the AMF; has been considered and distributed in Japan by BoFAS Japan, a registered securities dealer under the Financial Instruments and Exchange Act in Japan, or its permitted affiliates; is issued and distributed in Hong Kong by Merrill Lynch (Hong Kong) which is regulated by HKSF; is issued and distributed in Taiwan by Merrill Lynch (Taiwan); is issued and distributed in India by BoFAS India; and is issued and distributed in Singapore to institutional investors and/or accredited investors (each as defined under the Financial Advisers Regulations) by Merrill Lynch (Singapore) (Company Registration No 198602883D). Merrill Lynch (Singapore) is regulated by MAS. Merrill Lynch Equities (Australia) Limited (ABN 65 006 276 795), AFS License 235132 (MLEA) distributes this information in Australia only to 'Wholesale' clients as defined by s.761G of the Corporations Act 2001. With the exception of Bank of America N.A., Australia Branch, neither MLEA nor any of its affiliates involved in preparing this information is an Authorised Deposit-Taking Institution under the Banking Act 1959 nor regulated by the Australian Prudential Regulation Authority. No approval is required for publication or distribution of this information in Brazil and its local distribution is by Merrill Lynch (Brazil) in accordance with applicable regulations. Merrill Lynch (DIFC) is authorized and regulated by the DFSA. Information prepared and issued by Merrill Lynch (DIFC) is done so in accordance with the requirements of the DFSA conduct of business rules. BoFA Europe (Frankfurt) distributes this information in Germany and is regulated by BaFin, the ECB and the CBI. BoFA Securities entities, including BoFA Europe and BoFASE (France), may outsource/delegate the marketing and/or provision of certain research services or aspects of research services to other branches or members of the BoFA Securities group. You may be contacted by a different BoFA Securities entity acting for and on behalf of your service provider where permitted by applicable law. This does not change your service provider. Please refer to the [Electronic Communications Disclaimers \(http://www.bankofamerica.com/emaildisclaimer\)](http://www.bankofamerica.com/emaildisclaimer) for further information.

This information has been prepared and issued by BoFAS and/or one or more of its non-US affiliates. The author(s) of this information may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so. BoFAS and/or MLPF&S is the distributor of this information in the US and accepts full responsibility for information distributed to BoFAS and/or MLPF&S clients in the US by its non-US affiliates. Any US person receiving this information and wishing to effect any transaction in any security discussed herein should do so through BoFAS and/or MLPF&S and not such foreign affiliates. Hong Kong recipients of this information should contact Merrill Lynch (Asia Pacific) Limited in respect of any matters relating to dealing in securities or provision of specific advice on securities or any other matters arising from, or in connection with, this information. Singapore recipients of this information should contact Merrill Lynch (Singapore) Pte Ltd in respect of any matters arising from, or in connection with, this information. For clients that are not accredited investors, expert investors or institutional investors Merrill Lynch (Singapore) Pte Ltd accepts full responsibility for the contents of this information distributed to such clients in Singapore.

General Investment Related Disclosures:

Taiwan Readers: Neither the information nor any opinion expressed herein constitutes an offer or a solicitation of an offer to transact in any securities or other financial instrument. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or any other person without the express written consent of BoFA Securities.

This document provides general information only, and has been prepared for, and is intended for general distribution to, BoFA Securities clients. Neither the information nor any opinion expressed constitutes an offer or an invitation to make an offer, to buy or sell any securities or other financial instrument or any derivative related to such securities or instruments (e.g., options, futures, warrants, and contracts for differences). This document is not intended to provide personal investment advice and it does not take into account the specific investment objectives, financial situation and the particular needs of, and is not directed to, any specific person(s). This document and its content do not constitute, and should not be considered to constitute, investment advice for purposes of ERISA, the US tax code, the Investment Advisers Act or otherwise. Investors should seek financial advice regarding the appropriateness of investing in financial instruments and implementing investment strategies discussed or recommended in this document and should understand that statements regarding future prospects may not be realized. Any decision to purchase or subscribe for securities in any offering must be based solely on existing public information on such security or the information in the prospectus or other offering document issued in connection with such offering, and not on this document.

Securities and other financial instruments referred to herein, or recommended, offered or sold by BoFA Securities, are not insured by the Federal Deposit Insurance Corporation and are not deposits or other obligations of any insured depository institution (including, Bank of America, N.A.). Investments in general and, derivatives, in particular, involve numerous risks, including, among others, market risk, counterparty default risk and liquidity risk. No security, financial instrument or derivative is suitable for all investors. Digital assets are extremely speculative, volatile and are largely unregulated. In some cases, securities and other financial instruments may be difficult to value or sell and reliable information about the value or risks related to the security or financial instrument may be difficult to obtain. Investors should note that income from such securities and other financial instruments, if any, may fluctuate and that price or value of such securities and instruments may rise or fall and, in some cases, investors may lose their entire principal investment. Past performance is not necessarily a guide to future performance. Levels and basis for taxation may change.

This report may contain a short-term trading idea or recommendation, which highlights a specific near-term catalyst or event impacting the issuer or the market that is anticipated to have a short-term price impact on the equity securities of the issuer. Short-term trading ideas and recommendations are different from and do not affect a stock's fundamental equity rating, which reflects both a longer term total return expectation and attractiveness for investment relative to other stocks within its Coverage Cluster. Short-term trading ideas and recommendations may be more or less positive than a stock's fundamental equity rating.

BoFA Securities is aware that the implementation of the ideas expressed in this report may depend upon an investor's ability to "short" securities or other financial instruments and that such action may be limited by regulations prohibiting or restricting "shortselling" in many jurisdictions. Investors are urged to seek advice regarding the applicability of such regulations prior to executing any short idea contained in this report.

Foreign currency rates of exchange may adversely affect the value, price or income of any security or financial instrument mentioned herein. Investors in such securities and instruments, including ADRs, effectively assume currency risk.

BoFAS or one of its affiliates is a regular issuer of traded financial instruments linked to securities that may have been recommended in this report. BoFAS or one of its affiliates may, at any time, hold a trading position (long or short) in the securities and financial instruments discussed in this report.

BoFA Securities, through business units other than BoFA Global Research, may have issued and may in the future issue trading ideas or recommendations that are inconsistent with, and reach different conclusions from, the information presented herein. Such ideas or recommendations may reflect different time frames, assumptions, views and analytical methods of the persons who prepared them, and BoFA Securities is under no obligation to ensure that such other trading ideas or recommendations are brought to the attention of any recipient of this information. In the event that the recipient received this information pursuant to a contract between the recipient and BoFAS for the provision of research services for a separate fee, and in connection therewith BoFAS may be deemed to be acting as an investment adviser, such status relates, if at all, solely to the person with whom BoFAS has contracted directly and does not extend beyond the delivery of this report (unless otherwise agreed specifically in writing by BoFAS). If such recipient uses the services of BoFAS in connection with the sale or purchase of a security referred to herein, BoFAS may act as principal for its own account or as agent for another person. BoFAS is and continues to act solely as a broker-dealer in connection with the execution of any transactions, including transactions in any securities referred to herein.

Copyright and General Information:

Copyright 2024 Bank of America Corporation. All rights reserved. iQdatabase® is a registered service mark of Bank of America Corporation. This information is prepared for the use of BoFA Securities clients and may not be redistributed, retransmitted or disclosed, in whole or in part, or in any form or manner, without the express written consent of BoFA Securities. BoFA Global Research information is distributed simultaneously to internal and client websites and other portals by BoFA Securities and is not publicly-available material. Any unauthorized use or disclosure is prohibited. Receipt and review of this information constitutes your agreement not to redistribute, retransmit, or disclose to others the contents, opinions, conclusion, or information contained herein (including any investment recommendations, estimates or price targets) without first obtaining express permission from an authorized officer of BoFA Securities.

Materials prepared by BoFA Global Research personnel are based on public information. Facts and views presented in this material have not been reviewed by, and may not reflect information known to, professionals in other business areas of BoFA Securities, including investment banking personnel. BoFA Securities has established information barriers between BoFA Global Research and certain business groups. As a result, BoFA Securities does not disclose certain client relationships with, or compensation received from, such issuers. To the extent this material discusses any legal proceeding or issues, it has not been prepared as nor is it intended to express any legal conclusion, opinion or advice. Investors should consult their own legal advisers as to issues of law relating to the subject matter of this material. BoFA Global Research personnel's knowledge of legal proceedings in which any BoFA Securities entity and/or its directors, officers and employees may be plaintiffs, defendants, co-defendants or co-plaintiffs with or involving issuers mentioned in this material is based on public information. Facts and views presented in this material that relate to any such proceedings have not been reviewed by, discussed with, and may not reflect information known to, professionals in other business areas of BoFA Securities in connection with the legal proceedings or matters relevant to such proceedings.

This information has been prepared independently of any issuer of securities mentioned herein and not in connection with any proposed offering of securities or as agent of any issuer of any securities. None of BoFAS any of its affiliates or their research analysts has any authority whatsoever to make any representation or warranty on behalf of the issuer(s). BoFA Global Research policy prohibits research personnel from disclosing a recommendation, investment rating, or investment thesis for review by an issuer prior to the publication of a research report containing such rating, recommendation or investment thesis.

Any information relating to sustainability in this material is limited as discussed herein and is not intended to provide a comprehensive view on any sustainability claim with respect to any issuer or security.

Any information relating to the tax status of financial instruments discussed herein is not intended to provide tax advice or to be used by anyone to provide tax advice. Investors are urged to seek tax advice based on their particular circumstances from an independent tax professional.

The Information herein (other than disclosure information relating to BoFA Securities and its affiliates) was obtained from various sources and we do not guarantee its accuracy. This information may contain links to third-party websites. BoFA Securities is not responsible for the content of any third-party website or any linked content contained in a third-party website. Content contained on such third-party websites is not part of this information and is not incorporated by reference. The inclusion of a link does not imply any endorsement by or any affiliation with BoFA Securities. Access to any third-party website is at your own risk, and you should always review the terms and privacy policies at third-party websites before submitting any personal information to them. BoFA Securities is not responsible for such terms and privacy policies and expressly disclaims any liability for them.

All opinions, projections and estimates constitute the judgment of the author as of the date of publication and are subject to change without notice. Prices also are subject to change without notice. BoFA Securities is under no obligation to update this information and BoFA Securities ability to publish information on the subject issuer(s) in the future is subject to applicable quiet periods. You should therefore assume that BoFA Securities will not update any fact, circumstance or opinion contained herein.

Subject to the quiet period applicable under laws of the various jurisdictions in which we distribute research reports and other legal and BoFA Securities policy-related restrictions on the publication of research reports, fundamental equity reports are produced on a regular basis as necessary to keep the investment recommendation current.

Certain outstanding reports or investment opinions relating to securities, financial instruments and/or issuers may no longer be current. Always refer to the most recent research report relating to an issuer prior to making an investment decision.

In some cases, an issuer may be classified as Restricted or may be Under Review or Extended Review. In each case, investors should consider any investment opinion relating to such issuer (or its security and/or financial instruments) to be suspended or withdrawn and should not rely on the analyses and investment opinion(s) pertaining to such issuer (or its securities and/or financial instruments) nor should the analyses or opinion(s) be considered a solicitation of any kind. Sales persons and financial advisors affiliated with BoFAS or any of its affiliates may not solicit purchases of securities or financial instruments that are Restricted or Under Review and may only solicit securities under Extended Review in accordance with firm policies.

Neither BoFA Securities nor any officer or employee of BoFA Securities accepts any liability whatsoever for any direct, indirect or consequential damages or losses arising from any use of this information.

Terms of Use (<https://rsch.baml.com/WebReports/TermsOfUse.pdf>) | Privacy & Security: [GBAM \(https://www.bofam.com/en-us/content/global-privacy-notices.html\)](https://www.bofam.com/en-us/content/global-privacy-notices.html) [GWIM \(https://www.bankofamerica.com/privacy/privacy-overview.go\)](https://www.bankofamerica.com/privacy/privacy-overview.go) |

Cookie Policy (https://www.bofam.com/content/dam/boamlimages/documents/articles/ID19_0760/Cookie_Policy.pdf) | [GBAM Digital Privacy Notice \(https://business.bofa.com/content/dam/flagship/gbgrm/global-privacy-notices/BoFA-](https://business.bofa.com/content/dam/flagship/gbgrm/global-privacy-notices/BoFA-Digital-Privacy-Notice.pdf)

[Digital-Privacy-Notice.pdf](https://business.bofa.com/content/dam/flagship/gbgrm/global-privacy-notices/BoFA-Digital-Privacy-Notice.pdf)) | [GDPR Privacy Notice \(https://www.bofam.com/gdpr\)](https://www.bofam.com/gdpr) | [Your Privacy Choices](#)